CONNECTED COMPONENTS OF A FINITE BASE FIELD EXTENSION MAP ONTO CONNECTED COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma notes that it will be superseded by a stronger statement later in the chapter; that remark is reproduced outside the counted span and the stronger statement is counted separately by its own card.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-image-connected-component-finite-extension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- A scheme over a field is universally open over the base. Morphisms, Lemma morphisms-lemma-scheme-over-field-universally-open, invoked at proof lines 1136-1137. Supplies openness of the projection.
- An integral morphism is universally closed. Morphisms, Lemma morphisms-lemma-integral-universally-closed, invoked at proof lines 1138-1139. Supplies closedness of the projection, since a finite field extension is integral.
- A finite morphism is quasi-finite, hence has finite fibres. Morphisms, Lemma morphisms-lemma-finite-quasi-finite, invoked at proof lines 1140-1141. Supplies finiteness of the fibres of the projection.
- Behaviour of connected components under an open, closed map with finite fibres. Topology, Lemma topology-lemma-finite-fibre-connected-components, invoked at proof lines 1142-1144. Supplies the conclusion once the three properties are available.

Source reference. Original lemma 02NU: sources/stacks-chow/tags.txt, source lines 3493–3493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WM: sources/stacks-chow/tags.txt, source lines 2487–2487. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 07VB: sources/stacks-chow/tags.txt, source lines 10001–10001. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let K/k be a finite extension of fields and let X be a scheme over k. Denote by $p : X_K \to X$ the projection morphism. For every connected component T of X_K the image $p(T)$ is a connected component of X.*

Proof. The image $p(T)$ is contained in some connected component X' of X. Consider X' as a closed subscheme of X in any way. Then T is also a connected component of $X'_K = p^{-1}(X')$ and we may therefore assume that X is connected. The morphism p is open (Morphisms, Lemma 0383), closed (Morphisms, Lemma 01WM) and the fibers of p are finite sets (Morphisms, Lemma 02NU). Thus we may apply Topology, Lemma 07VB to conclude. □